SRS – Financial Referral CRM System

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1. Project Overview & Business Flow

1.1 Project Overview

The Financial Referral CRM is a web-based investment and referral management system designed to automate profit-sharing, track user investments, and manage a multi-level referral structure.

The system supports:

- Investment tracking per user
- Semi-monthly profit sharing (14th & last day of every month)
- Referral-based income distribution
- Automated commission calculations
- Admin-controlled financial operations

The platform ensures transparency, traceability, and automation of financial flows.

1.2 Business Flow

Step 1: User Registration

- User signs up using email/phone
- Optional referral code applied
- User account is created under referral tree

Step 2: Investment Creation

- User invests on any date
- Investment is recorded with:
    - Amount
    - Start date
    - Active status
- Investment becomes eligible for profit sharing immediately

Step 3: Profit Cycle Activation

Profit is calculated twice every month:

- **1st Cycle:** 1st to 14th → payout on 14th
- **2nd Cycle:** 15th to end of month → payout on last day

Each investment earns profit based on:

- Number of active days in cycle
- Defined profit-sharing percentage

Step 4: Referral Tracking

- Each user has a referral tree
- Referrers earn commission based on:
  - Direct referrals
  - Possibly multi-level structure (if enabled)

Step 5: Commission Distribution

- System calculates:
  - Profit share
  - Referral commission
  - Admin deductions (if any)
- Wallets are updated automatically

Step 6: Withdrawal (Optional Module)

- Users request withdrawal from wallet
- Admin approves/rejects
- Transaction recorded

Step 7: Reporting & Audit

- All transactions logged
- Admin can view full financial history
- Reports generated per cycle

1.3 Core Objective

To automate investment-based profit distribution with strict rule-based financial calculations and referral-based earning system.

2. Functional Requirements

This section defines all system capabilities required to implement the Financial Referral CRM.

2.1 User Management Module

Features

- User registration with email/phone
- Login/logout authentication
- Password reset
- Profile management
- Referral code assignment per user
- View referral tree (basic level visibility)

User Types

- Super Admin
- Admin
- User

2.2 Investment Management Module

Features

- Create investment entry
- Track investment status (Active / Completed / Suspended)
- Store investment start date and amount
- Multiple investments per user allowed
- Investment history per user

Investment Fields

- Investment ID

- User ID
- Amount
- Start Date
- Status
- Created Timestamp

2.3 Profit Sharing Module

Features

- Automatic calculation every cycle
- Two payout cycles per month:
  - 1st–14th (payout on 14th)
  - 15th–end of month (payout on last day)
- Per-day profit calculation
- Allocation based on active investment days
- Wallet crediting system

2.4 Referral Management Module

Features

- Unique referral code generation per user
- Track direct referrals
- Build referral tree structure
- Assign referral income rules
- Prevent self-referral abuse

2.5 Commission Module

Features

- Calculate referral commission
- Multi-level commission support (configurable)
- Credit commission to wallet
- Maintain commission history logs

2.6 Wallet Module

Features

- User wallet balance
- Separate ledger for:
  - Investment profit
  - Referral commission
  - Adjustments
- Transaction history
- Freeze/unfreeze wallet (admin control)

2.7 Withdrawal Module (Optional but Recommended)

Features

- Withdrawal request creation
- Admin approval workflow
- Status tracking (Pending / Approved / Rejected)
- Deduction from wallet after approval
- Transaction logs

2.8 Admin Control Module

Features

- Manage users
- Manage investments
- Approve withdrawals
- Configure profit percentage
- Configure referral commission rates
- View system reports

2.9 Reporting Module

Features

- Daily / Weekly / Monthly reports
- Investment summary reports

- Profit distribution reports
- Referral earnings reports
- Export reports (CSV/PDF)

2.10 Notification Module

Features

- Email notifications
- SMS notifications (optional)
- In-app notifications
- Alerts for:
  - Investment confirmation
  - Profit credited
  - Withdrawal status
  - Referral earnings

3. Super Admin Portal (Revised – Core Control System)

The Super Admin Portal is the central authority system of the Financial Referral CRM. It is designed to provide complete operational, financial, and administrative control over the entire platform.

This portal eliminates wallet-based complexity and focuses on **direct payout execution, structured profit calculations, and system-level reporting**.

3.1 Super Admin Dashboard

Overview Metrics

- Total Users
- Active Investments
- Total Investment Volume
- Total Profit Generated (Cycle-wise)
- Total Payout Released
- Pending Payout Amount (System Liability)
- Active Admin Accounts

Live System Activity

- Recent user registrations
- Recent investments
- Payout release history
- Admin activity logs

3.2 User Management (Full CRUD Control)

Super Admin has complete control over all user accounts.

Features

- Create new user manually
- Edit user details
- Delete user permanently or soft-delete
- Block / unblock users
- Assign referral codes manually
- Override referral assignments

User Profile View Includes

- Personal details
- Referral hierarchy tree
- Investment history
- Profit history
- Account status

3.3 Investment Management Module

Features

- View all investments in system
- Create / edit / delete investments
- Activate / deactivate investments
- Adjust investment start dates (admin override)
- Audit all investment changes

Controls

- Lock investments after activation
- Modify profit eligibility rules per investment (admin override only)

3.4 Profit Sharing Engine

This is the core financial logic system.

Features

- Auto-calculation of profit in two cycles:
  - 1st–14th → payout on 14th
  - 15th–end → payout on last day
- Per-day profit computation based on active investments
- System-wide profit generation report

Super Admin Controls

- Configure profit percentage
- Manually trigger profit calculation
- Recalculate previous cycles
- Adjust profit rules globally

3.5 One-Click Payout System (NO WALLET MODEL)

Instead of wallets, the system uses **direct payout release mechanism**.

Features

- Generate payout list per cycle
- One-click “Release All Payments”
- Individual payout release option
- Bulk payout by filters:
  - Date
  - User
  - Investment batch

Payout Flow

1. System calculates profit
2. Super Admin reviews payout sheet
3. Admin clicks:
    - **“Release All Payments”**
4. System marks all payouts as completed

Audit Requirement

- Every payout stored with:
    - User ID
    - Amount
    - Cycle date
    - Admin who released it

3.6 Referral Management System

Features

- Create referral structure manually or automatically
- View full referral tree (global visualization)
- Edit referral assignments
- Delete invalid referral links

Controls

- Enable/disable referral program globally
- Set referral depth (1-level / multi-level)
- Override referral commission rules

3.7 Admin Role & Permission Management

Super Admin controls all admin accounts and access levels.

Features

- Create admin accounts
- Assign roles
- Modify permissions anytime
- Delete or deactivate admin accounts

Permission Categories

- User Management
- Investment Management
- Profit Engine Access
- Payout Release Permission
- Report Access
- System Settings Access

Example Roles

- Finance Admin (can release payouts)
- Operations Admin (can manage users/investments)
- Analyst Admin (read-only reports)

3.8 Reports Module (Full System Reporting)

Available Reports

- User registration report
- Investment report
- Profit generation report
- Payout report
- Referral performance report
- Admin activity report

Features

- Date range filtering
- User-wise filtering
- Export formats:
  - PDF
  - Excel
  - CSV

3.9 System Configuration Panel

Core Settings

- Profit percentage configuration
- Cycle schedule settings
- Referral rules configuration
- System timezone
- Auto-calculation scheduler ON/OFF

Financial Rules Control

- Profit formula configuration
- Minimum investment rules (optional)
- Manual override permissions

3.10 Audit & Activity Logs

Tracked Actions

- User creation/edit/deletion
- Investment changes
- Profit calculations
- Payout releases
- Admin login history

Security Logs

- IP tracking
- Device tracking
- Failed login attempts
- Suspicious activity alerts

3.11 Security Permissions Matrix

Module	Create	Read	Edit	Delete	Approve
Users	✔	✔	✔	✔	✔
Investments	✔	✔	✔	✔	✔
Profit Engine	✔	✔	✔	✔	✔
Payout System	✔	✔	✔	✔	✔

Module	Create	Read	Edit	Delete	Approve
Admin Roles	✔	✔	✔	✔	✔
Reports	✔	✔	✔	✔	✔

4. Admin Portal (Operational Management Layer)

The Admin Portal is a controlled operational interface designed for day-to-day system handling. Unlike the Super Admin, Admins have **restricted access based on permissions assigned by Super Admin**.

Admins do not control core financial logic but assist in managing users, investments, and monitoring system activity.

4.1 Purpose of Admin Portal

The Admin Portal is designed to:

- Support Super Admin operations
- Manage users and investments (based on permission)
- Monitor system activity
- Generate operational reports
- Assist in verifying data before payout execution

4.2 Admin Dashboard

Overview Metrics

- Assigned Users Count
- Active Investments (under assigned scope)
- Total Investment Value (scope-based)
- Pending Actions (if any)
- Recent User Activity

Activity Feed

- New user registrations
- New investments
- Referral updates
- System notifications

4.3 User Management (Permission-Based)

Allowed Actions (if permitted)

- View users
- Edit user details
- Block / unblock users
- View referral details

Restricted Actions

- ❌ Create user (optional restriction)
- ❌ Delete user (Super Admin only, unless enabled)
- ❌ Modify referral structure globally

User View Includes

- Profile information
- Referral chain
- Investment summary
- Activity history

4.4 Investment Management Module

Features

- View investments (based on scope)
- Create investments (if permission enabled)
- Update investment status
- View investment history
- Export investment lists

Controls

- Mark investment as verified
- Flag suspicious investments (for Super Admin review)
- Edit investment metadata (limited fields)

4.5 Profit Monitoring Module (Read-Only / Controlled Access)

Admins cannot modify profit rules but can monitor calculations.

Features

- View profit cycle reports
- View calculated earnings per investment
- Track payout readiness status
- Identify discrepancies

Access Level

- Read-only for most admins
- Finance admins may have extended visibility

4.6 Payout Review Module (Pre-Approval Stage)

Admins act as reviewers before Super Admin releases payments.

Features

- View payout batches
- Validate payout calculations
- Flag incorrect entries
- Approve payout sheet for Super Admin

Workflow

1. System generates payout batch
2. Admin reviews entries
3. Admin marks as “Verified”

4. Super Admin releases payments

4.7 Referral Monitoring Module

Features

- View referral tree (limited depth access)
- Track referral activity
- Monitor referral earnings distribution
- Detect invalid referral patterns

Restrictions

- ❌ Cannot modify referral hierarchy
- ❌ Cannot delete referral links

4.8 Reports Module (Scoped Access)

Admins can generate reports based on assigned permissions.

Available Reports

- Assigned user reports
- Investment reports
- Cycle-based profit reports
- Referral activity reports

Export Options

- PDF
- Excel
- CSV

4.9 Notification Center

Features

- System alerts
- Investment updates
- Payout cycle notifications
- Admin task assignments

Types

- System-generated
- Super Admin messages
- Audit alerts

4.10 Admin Permission Model

Permissions are fully controlled by Super Admin.

Permission Categories

Module	View	Create	Edit	Delete	Approve
Users	✔	Optional	✔	✕	✕
Investments	✔	Optional	✔	✕	✔
Profit Data	✔	✕	✕	✕	✕
Payout Review	✔	✕	✕	✕	✔
Reports	✔	✕	✕	✕	✕

4.11 Admin Activity Logs

Tracked Actions

- User edits
- Investment updates
- Report generation
- Login/logout activity
- Payout review actions

Security Monitoring

- IP tracking
- Device tracking
- Failed login attempts

- Suspicious activity alerts

4.12 Role Types (Example Structure)

- Finance Admin → payout verification + reports
- Operations Admin → user + investment handling
- Support Admin → user assistance only
- Analyst Admin → read-only dashboards

5. User Portal (Investor Dashboard System)

The User Portal is the front-facing dashboard for investors. It allows users to track investments, view profit earnings, monitor referral performance, and manage their account activity.

The system is designed to be **read + track oriented**, not administrative. Users cannot modify financial rules or system data.

5.1 Purpose of User Portal

The User Portal is designed to:

- Provide transparency of investments and returns
- Show real-time profit calculations
- Display referral earnings and structure
- Allow users to track financial growth
- Maintain full audit visibility of transactions

5.2 User Dashboard (Home Screen)

Overview Metrics

- Total Investment Amount
- Active Investments Count
- Total Profit Earned
- Total Referral Earnings
- Current Cycle Status (1–14 / 15–End)
- Next Payout Date (Display only)

Live Activity Panel

- Latest profit updates
- Referral activity notifications
- Investment status changes
- System announcements

5.3 Investment Tracking Module

Features

- View all investments
- Investment status tracking:
    - Active
    - Completed
    - Suspended (if applicable)
- Investment start date and cycle mapping
- Daily profit visibility (calculated view)

Investment Detail View

Each investment shows:

- Investment ID
- Amount invested
- Start date
- Active days in current cycle
- Profit earned per cycle
- Total profit till date

5.4 Profit Visibility Module

Features

- Cycle-wise profit display:
    - Cycle 1: 1–14
    - Cycle 2: 15–end of month
- Total profit summary
- Daily earning breakdown (calculated view)
- Historical profit records

Display Format

- Current cycle earnings
- Previous cycle earnings
- Lifetime profit summary

5.5 Referral Dashboard

Features

- Unique referral code display
- Referral link sharing option
- Direct referral count
- Referral earnings summary

Referral Tree View

- Visual hierarchy structure
- Level-based display (if multi-level enabled)
- Direct + indirect referral tracking

Referral Stats

- Total referrals
- Active referrals
- Earnings from referrals

5.6 Transaction History Module

Features

- Complete financial history
- Investment entries
- Profit credits
- Referral earnings

Filters

- Date range

- Transaction type
- Cycle-based filtering

5.7 Notification Center

Features

- Profit credited alerts
- Investment confirmations
- Referral earnings notifications
- System announcements
- Admin messages (read-only)

5.8 Profile Management

Features

- View personal details
- Update contact information
- Change password
- View referral code
- Account status

Restrictions

- ❌ Cannot modify investments
- ❌ Cannot change referral structure
- ❌ Cannot alter financial data

5.9 Earnings Summary Module

Breakdown

- Investment profit earnings
- Referral commission earnings
- Total earnings (combined view)

Visualization

- Cycle-wise chart (optional)
- Monthly earnings summary
- Growth trend indicators

5.10 Security & Access Control

Features

- Login session tracking
- Device recognition
- IP logging
- Logout from all devices option

Protection Rules

- Multi-session monitoring
- Failed login lockout system
- OTP verification (optional enhancement)

5.11 User Limitations

Users are strictly restricted from:

- Editing investments
- Modifying profit rules
- Changing referral hierarchy
- Accessing admin tools
- Initiating payouts or financial releases

5.12 User Experience Flow

1. Login → Dashboard
2. View investments → Track profit
3. Monitor referrals → Check earnings
4. Wait for cycle updates (14th / End of month)

5. View updated profit credits automatically

6. Business Rules Engine (Core Logic Layer)

The Business Rules Engine is the **heart of the Financial Referral CRM system**. It defines how investments behave, how profits are calculated, and how cycles are executed in a deterministic, auditable way.

This engine ensures that every financial output is **rule-based, time-bound, and fully traceable**.

6.1 Core Purpose

The Business Rules Engine is responsible for:

- Defining profit calculation logic
- Managing investment eligibility
- Controlling cycle-based processing (14th & month-end)
- Enforcing system constraints
- Handling edge cases and exceptions
- Ensuring consistent financial outputs across the system

6.2 System Time Structure (Critical Foundation)

The entire system is based on **monthly bi-cycle segmentation**:

Cycle Structure

Cycle 1

- Start: 1st of month
- End: 14th of month
- Payout Date: 14th

Cycle 2

- Start: 15th of month
- End: Last day of month
- Payout Date: Last day of month

System Time Rule

- All calculations are time-zone locked (system default)
- No retroactive cycle shifting unless admin override is used

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6.3 Profit Calculation Core Logic

6.3.1 Base Formula

```
Profit = Investment Amount × Daily Profit Rate × Active Days in Cycle
```

6.3.2 Daily Profit Rate

Defined at system level:

- Example:
  - Monthly ROI = 10%
  - Daily Rate = 10% ÷ 30 (or actual month days logic)

6.3.3 Active Days Calculation Rule

An investment earns profit only for days it is active within a cycle.

Example

- Investment date: 5th
- Cycle: 1–14

Active days = 5th → 14th = 10 days

6.3.4 Partial Cycle Logic

If investment starts mid-cycle:

- Profit is prorated from start date
- No backdated earnings

If investment ends or is deactivated:

- Profit stops immediately

6.4 Investment Eligibility Rules

Eligible Conditions

- Investment must be ACTIVE
- Must fall within a valid cycle window
- Must not be suspended or deleted
- Must have valid start date

Ineligible Conditions

- Future-dated investments (unless allowed by admin)
- Suspended investments
- Deleted or soft-removed records
- Manually flagged fraudulent investments

6.5 Cycle Processing Rules

6.5.1 Automatic Cycle Trigger

System automatically triggers calculations on:

- 14th of every month
- Last day of every month

6.5.2 Cycle Execution Flow

1. Lock cycle data (freeze inputs)
2. Fetch all active investments
3. Calculate per-investment profit
4. Generate payout batch
5. Send batch to Super Admin approval/release

6.5.3 Cycle Lock Rule

Once a cycle is closed:

- No changes allowed to past calculations
- Only Super Admin override can reopen cycle

6.6 Constraints & System Limits

Financial Constraints

- No negative profit allowed
- No duplicate payout entries
- No overlapping cycle payouts
- Each investment can only be processed once per cycle

Data Constraints

- Investment must always have:
  - User ID
  - Start date
  - Amount
- Missing fields = invalid record

Operational Constraints

- Payout cannot be released before cycle completion
- Admin cannot bypass calculation engine
- Super Admin override is logged and audited

6.7 Referral Interaction Rules (Linked Logic)

While profit engine is separate, it can interact with referral logic:

- Referral earnings do NOT affect investment profit
- Referral commissions are calculated independently
- No compounding between referral + investment profit unless enabled

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6.8 Edge Cases Handling

6.8.1 Mid-Cycle Investment

If user invests on:

- 5th → earns from 5th to cycle end
- 15th → only second cycle applies

6.8.2 Same-Day Investment & Payout

If investment happens on payout day:

- System rule: depends on cut-off time
- Default: included in next cycle

6.8.3 Month Length Variation

System adapts automatically:

- 28 / 29 / 30 / 31 day months supported
- No manual adjustment needed

6.8.4 System Downtime During Cycle

If system is offline during payout day:

- Calculations queued
- Executed when system resumes
- No data loss allowed

6.8.5 Admin Override Scenarios

Allowed only for:

- Incorrect calculation correction
- Fraud adjustment
- Emergency recalculation

Every override must be:

- Logged
- Timestamped
- Assigned to admin ID

6.8.6 Deleted Investment Handling

- If deleted before cycle → ignored completely
- If deleted after cycle → remains in historical audit only

6.8.7 Duplicate Investment Prevention

System prevents:

- Same user + same timestamp duplication
- Duplicate transaction IDs

6.9 System Determinism Rule

The engine must always produce:

- Same input → same output
- No randomness in financial calculations
- Fully reproducible audit trail

6.10 Output of Business Rules Engine

The engine produces:

- Cycle-wise profit sheets
- Investment-wise breakdown
- Payout batch data
- Audit logs
- Exception reports

7. Profit Sharing Engine (Technical Design & Execution Layer)

The Profit Sharing Engine is the execution system that applies the Business Rules Engine into real financial calculations. It processes investments, calculates cycle-based earnings, and generates payout-ready batches for Super Admin approval.

This engine is designed for **accuracy, scalability, and auditability**.

7.1 Purpose of the Profit Sharing Engine

The engine is responsible for:

- Processing all active investments per cycle
- Calculating daily profit earnings
- Generating cycle-wise payout batches
- Ensuring consistency with business rules
- Maintaining full audit trails for every calculation

7.2 System Architecture

Core Components

- **Scheduler Service** → triggers cycle execution
- **Calculation Engine** → computes profit per investment
- **Batch Generator** → groups payout data
- **Audit Logger** → records all computations
- **Validation Layer** → ensures data integrity

7.3 Cycle Execution Workflow

Trigger Points

- 14th of every month
- Last day of every month

Execution Steps

Step 1: Cycle Initialization

- Lock cycle window
- Freeze investment dataset
- Capture system timestamp snapshot

Step 2: Data Fetching

Retrieve:

- All active investments
- Valid user accounts
- System profit rate configuration

Step 3: Eligibility Filtering

Filter out:

- Suspended investments
- Invalid records
- Deleted accounts
- Out-of-cycle investments

Step 4: Profit Calculation Engine Execution

Each investment is processed individually:

```
Profit = Investment Amount × Daily Rate × Active Days in Cycle
```

Processing Logic

- Determine cycle window (1–14 or 15–end)
- Calculate overlap with investment active period
- Compute eligible days
- Apply daily profit rate

Step 5: Aggregation Layer

- Sum all investment profits per user
- Combine cycle-wise earnings
- Prepare payout-ready dataset

Step 6: Batch Generation

System generates:

Payout Batch Structure

- Batch ID
- Cycle period
- User-wise breakdown
- Total payout amount
- Timestamp

Step 7: Audit Logging

Every calculation is stored:

- Investment ID
- User ID
- Active days
- Rate applied
- Final profit
- Cycle ID
- Engine version

7.4 Calculation Optimization Strategy

To handle large datasets:

Batch Processing

- Process users in chunks (e.g., 500–1000 records)
- Parallel computation enabled

Caching Layer

- Store system rates in memory
- Avoid repeated DB reads

Pre-aggregation

- Group investments by user before calculation

7.5 Profit Distribution Logic (Output Stage)

After calculation:

Output Format

- User ID
- Total cycle profit
- Breakdown per investment
- Cycle ID
- Status = “Ready for Release”

Important Rule

No direct payout is executed by this engine.

It only produces:

Verified payout batches for Super Admin release

7.6 Recalculation Engine

When Triggered

- Admin correction request
- System rule updates
- Audit mismatch detection

Rules

- Old batch is invalidated (soft versioning)
- New batch is generated
- Full comparison log stored

7.7 Edge Case Handling

7.7.1 Zero-Day Investment

- If investment starts and ends outside cycle → profit = 0

7.7.2 Partial Overlap Investment

Example:

- Investment starts 10th
- Cycle 1 (1–14)

Only days 10–14 are counted

7.7.3 System Delay During Cycle

- Calculation is queued
- Executed on restart
- Cycle timestamp remains unchanged

7.7.4 Multiple Investments per User

- Each investment calculated independently

- Aggregation happens at user level

7.7.5 Rate Change During Cycle

Rule:

- Rate is locked at cycle start
- Mid-cycle changes apply only to next cycle

7.8 Data Integrity Rules

- No duplicate calculations per cycle
- Each investment processed once per cycle
- Immutable audit logs
- Hash-based batch validation (optional enhancement)

7.9 Engine Output Types

Primary Outputs

- Cycle payout batch
- User earnings report
- Investment breakdown report

Secondary Outputs

- Audit logs
- Exception reports
- System performance logs

7.10 Performance Expectations

- Supports 10,000+ investments per cycle
- Calculation completion under minutes (scalable)
- Horizontal scaling supported via worker nodes

7.11 System Reliability Rules

- Fail-safe retry mechanism for failed calculations
- Transaction rollback support
- No partial payout generation allowed

8. Referral Tree Engine (Multi-Level Structure & Commission Flow System)

The Referral Tree Engine is responsible for building, maintaining, and propagating the entire **hierarchical referral network** of the Financial Referral CRM. It also governs how referral commissions are calculated and distributed across levels.

This engine ensures a **structured genealogy model + deterministic commission flow + auditable referral mapping**.

8.1 Purpose of Referral Tree Engine

The engine is designed to:

- Build multi-level referral hierarchy
- Track user-to-user relationships
- Propagate referral chains automatically
- Calculate referral commissions
- Maintain tree integrity and prevent manipulation
- Support visualization of referral networks

8.2 Referral Tree Structure

8.2.1 Base Model (Graph Structure)

Each user is represented as a **node** in a directed tree:

- Each user has:
  - One parent (referrer)
  - Multiple children (direct referrals)

Structure Example

```
User A (Root)
├── User B
│    ├── User D
│    └── User E
└── User C
     ├── User F
     └── User G
```

8.3 Multi-Level Propagation Logic

8.3.1 Referral Depth Rules

System supports configurable depth:

- Level 1 → Direct referral
- Level 2 → Referral of referral
- Level 3+ → Optional (configurable)

8.3.2 Propagation Rule

When a new user joins:

1. System checks referral code
2. Assigns parent node
3. Propagates lineage upward
4. Stores full ancestry path

8.3.3 Immutable Chain Rule

Once assigned:

- Referral hierarchy cannot be changed (except Super Admin override)
- Prevents retroactive commission manipulation

8.4 Commission Engine Logic

Referral commissions are calculated independently of investment profit.

8.4.1 Commission Formula

```
Commission = Downline Investment × Commission Rate(Level)
```

8.4.2 Level-Based Commission Structure

Example configuration:

Level	Commission %
Level 1	5%
Level 2	2%
Level 3	1%

8.4.3 Commission Trigger Events

Commissions are generated when:

- New investment is created
- Investment becomes active
- System cycle is processed (optional rule)

8.4.4 Commission Flow Path

When User D invests:

```
User D invests
→ User B gets Level 1 commission
→ User A gets Level 2 commission
```

8.5 Tree Traversal Logic

8.5.1 Upward Traversal

Used for commission calculation:

- Start from investing user
- Move upward through parent nodes
- Stop at configured max depth

8.5.2 Downward Traversal

Used for visualization:

- Start from root user
- Expand all child nodes recursively

8.5.3 Breadth vs Depth Control

- BFS (Breadth-first search) → For tree visualization
- DFS (Depth-first search) → For commission propagation

8.6 Referral Validation Rules

Valid Referral Conditions

- Referral code must exist
- Referrer must be active
- No circular references allowed

Invalid Conditions

- Self-referral attempt → rejected
- Invalid referral code → null parent
- Circular tree detection → system block

8.7 Tree Integrity Constraints

Hard Rules

- One user = one parent only
- No loops allowed in hierarchy
- No duplicate referral assignments
- Referral structure is append-only

Audit Enforcement

Every change logged with:

- User ID
- Parent ID
- Admin ID (if override)
- Timestamp
- Reason for modification

8.8 Commission Ledger System

Even though there is no wallet system, commissions are tracked as **ledger entries**:

Ledger Entry Includes

- Recipient User ID
- Source User ID (downline investor)
- Investment ID
- Level (1, 2, 3…)
- Commission amount
- Timestamp
- Cycle reference

8.9 Performance Optimization Strategy

8.9.1 Tree Indexing

- Materialized path storage
- Cached parent-child relationships

8.9.2 Batch Commission Calculation

- Process commissions in bulk per investment batch
- Avoid per-user recursion at runtime

8.9.3 Precomputed Hierarchy Cache

- Store referral chain snapshot for each user
- Reduces traversal cost during payout cycles

8.10 Tree Visualization Model

8.10.1 Visualization Format

The UI displays referral structure as:

- Node-based tree
- Expand/collapse hierarchy
- Level-based coloring (optional)

8.10.2 Data Format for UI

Each node contains:

- User ID
- Name
- Level depth
- Direct referrals count
- Total downline count
- Investment summary (optional)

8.10.3 Example JSON Structure

```
{
  "userId": "U001",
```

```
  "name": "User A",
  "children": [
    {
      "userId": "U002",
      "name": "User B",
      "children": []
    }
  ]
}
```

8.10.4 Visualization Rules

- Maximum depth default: 3–5 levels
- Lazy loading for large trees
- Expand on click for performance

8.11 Edge Cases Handling

8.11.1 Missing Referrer

- Assigned to system root node

8.11.2 Deleted User in Tree

- Node remains in history (soft delete)
- Children remain attached unless reassigned

8.11.3 Circular Reference Attempt

- System detects loop
- Blocks assignment automatically

8.11.4 Massive Downline Trees

- Pagination or lazy loading applied
- API throttling enabled

8.12 Output of Referral Engine

The engine produces:

- Referral tree structure
- Commission ledger entries
- Level-wise earnings data
- Tree visualization dataset

9. Commission Engine (Rules, Calculation Logic & Reconciliation System)

The Commission Engine is a dedicated financial subsystem responsible for calculating, validating, and recording all referral-based commissions generated through investments.

It operates independently from the Profit Sharing Engine to ensure **clear separation between investment returns and referral earnings**.

9.1 Purpose of Commission Engine

The engine is designed to:

- Calculate referral commissions based on investment activity
- Apply multi-level commission rules
- Maintain a structured commission ledger
- Prevent duplication or overpayment
- Reconcile commissions per cycle or real-time events

9.2 Core Commission Principles

Separation Rule

- Investment profit ≠ Referral commission
- Both systems operate independently

Event-Based System

Commissions are triggered by:

- New investment creation
- Investment activation
- Optional cycle-based recalculation

9.3 Commission Calculation Logic

9.3.1 Base Formula

```
Commission = Investment Amount × Level Rate
```

9.3.2 Multi-Level Calculation Flow

When a user invests, commission flows upward:

Example:

User D invests ₹10,000

Level	User	Rate	Commission
L1	User B	5%	₹500
L2	User A	2%	₹200

9.3.3 Configurable Rate Engine

Rates are defined at system level:

- Level 1: X%
- Level 2: Y%
- Level 3: Z%

Rules can be:

- Global
- Admin-configurable
- Time-based (optional future enhancement)

9.4 Commission Trigger System

Triggers

Commission is generated when:

1. Investment Created

- Immediate commission calculation

2. Investment Activated

- Optional recalculation trigger

3. Cycle Processing (Optional Mode)

- Batch recalculation for audit consistency

Trigger Priority Order

1. Investment creation (primary)
2. System validation layer
3. Cycle reconciliation (secondary)

9.5 Commission Ledger System

Every commission is stored as an immutable ledger entry.

Ledger Structure

- Commission ID
- Recipient User ID
- Source User ID
- Investment ID
- Level (1/2/3…)
- Commission Amount
- Timestamp
- Status (Pending / Confirmed / Reconciled)

Immutability Rule

- Once recorded → cannot be modified
- Adjustments only via reversal entry

9.6 Commission Aggregation Logic

User-Level Aggregation

All commissions are grouped per user:

- Daily total
- Cycle total
- Lifetime total

Batch Aggregation

System also produces:

- Cycle-wise commission batch
- Investment-wise commission breakdown

9.7 Reconciliation Engine

Purpose

Ensures accuracy between:

- Investment records
- Referral tree
- Commission ledger

Reconciliation Process

1. Recalculate expected commissions

2. Compare with ledger entries
3. Identify mismatches:
    - Over-credit
    - Missing entries
    - Duplicate entries
4. Generate correction report

Correction Method

- No direct edits allowed
- System generates:
    - Credit adjustment entry
    - Debit reversal entry

9.8 Edge Case Handling

9.8.1 Self-Referral Attempt

- Automatically blocked
- No commission generated

9.8.2 Missing Parent Node

- Commission skipped
- Logged as "unassigned referral path"

9.8.3 Circular Referral Detection

- System validates tree integrity before processing
- Blocks execution if loop detected

9.8.4 Duplicate Investment Event

- Same investment ID cannot generate commission twice

9.8.5 Late Correction Scenario

If referral tree is corrected after commission:

- Old commission remains
- New correction entry is added (no overwrite)

9.9 Commission Status Lifecycle

Each commission goes through:

1. Pending
2. Calculated
3. Confirmed
4. Reconciled

9.10 Performance Optimization

Batch Processing

- Process commissions in bulk per investment batch
- Avoid per-user recursive calculation in runtime

Caching Strategy

- Cache referral hierarchy
- Cache commission rate tables

Precomputed Paths

- Store full referral chain per user
- Speeds up upward traversal

9.11 System Constraints

- No negative commission allowed
- No duplicate commission per investment-event-level
- Max depth enforced (configurable)
- Commission rate changes apply only to future events

9.12 Output of Commission Engine

The engine produces:

- Commission ledger entries
- User-wise commission summary
- Cycle-based commission reports
- Reconciliation audit reports
- Exception/error reports

9.13 Integration Points

Commission Engine connects with:

- Referral Tree Engine (hierarchy source)
- Investment Module (trigger source)
- Reporting Module (output layer)
- Super Admin Panel (review & override visibility)

10. Reports Module (Analytics, Financial Reports & Audit System)

The Reports Module is the centralized intelligence layer of the Financial Referral CRM. It consolidates data from investments, profit engine, referral system, and commission engine into structured, exportable, and audit-ready reports.

This module is critical for **financial transparency, regulatory compliance, and decision-making**.

10.1 Purpose of Reports Module

The system is designed to:

- Provide real-time and historical financial insights
- Offer cycle-wise and user-wise reporting
- Support audit and compliance requirements
- Enable Super Admin and Admin-level analytics
- Export structured financial data

10.2 Report Categories Overview

The system supports the following core report types:

1. User Reports
2. Investment Reports
3. Profit Sharing Reports
4. Referral Reports
5. Commission Reports
6. Payout Reports
7. Admin Activity Reports
8. Audit & System Logs Reports

10.3 User Reports Module

Purpose

Provides full visibility into user activity and lifecycle.

Includes

- User registration date
- Active / inactive status
- Total investments per user
- Total profit earned
- Total referral earnings
- Account activity timeline

Filters

- Date range
- User ID
- Status (Active / Blocked)

10.4 Investment Reports Module

Purpose

Tracks all investment activity across the system.

Includes

- Investment ID
- User ID
- Investment amount
- Start date
- Active status
- Cycle participation
- Total profit generated per investment

Breakdowns

- Daily investment inflow
- Monthly investment volume
- Active capital in system

10.5 Profit Sharing Reports Module

Purpose

Displays cycle-based profit distribution data.

Includes

- Cycle period (1–14 / 15–End)
- Total profit generated
- User-wise profit breakdown
- Investment-wise profit breakdown

Key Insights

- Total payout liability per cycle
- Average return per investment
- Profit distribution trends

10.6 Referral Reports Module

Purpose

Analyzes referral network performance.

Includes

- Referral tree depth analysis
- Total direct referrals per user
- Active referral count
- Referral-based investment inflow

Hierarchy Insights

- Top referrers ranking
- Level-wise contribution analysis
- Referral network growth chart

10.7 Commission Reports Module

Purpose

Tracks all referral commission activities.

Includes

- Commission ledger entries
- Level-wise commission distribution
- User-wise commission earnings
- Investment-linked commission mapping

Analytics

- Highest earning referrers
- Commission distribution ratio
- Average commission per investment

10.8 Payout Reports Module

(Even though there is no wallet system, payouts are still tracked as execution records.)

Purpose

Tracks all system payout executions.

Includes

- Payout batch ID
- Cycle reference
- Total payout amount
- User-wise payout breakdown
- Timestamp of release
- Admin who executed payout

Status Types

- Pending
- Generated
- Approved
- Released

10.9 Admin Activity Reports Module

Purpose

Tracks administrative actions across the system.

Includes

- Admin login history

- User modifications
- Investment changes
- Payout executions
- System configuration changes

Security Insights

- IP tracking
- Device logs
- Suspicious actions detection

10.10 Audit & System Logs Module

Purpose

Provides full traceability for compliance and debugging.

Includes

- System-generated logs
- Business rule execution logs
- Profit engine logs
- Commission engine logs
- Referral tree modifications

Immutability Rule

- Audit logs are read-only
- Cannot be edited or deleted (except system cleanup policy)

10.11 Export Formats

All reports support export in multiple formats:

Supported Formats

- PDF (formatted reports)
- Excel (.xlsx) (data analysis)
- CSV (raw data export)

- JSON (API integration)

Export Features

- Date range filtering before export
- User-specific export
- Cycle-specific export
- Bulk export support

10.12 Analytics Dashboard Layer

10.12.1 System-Wide Analytics

- Total investments over time
- Profit distribution trends
- Referral growth trends
- Commission payout trends

10.12.2 Financial Analytics

- Total capital inflow
- Total profit liability
- Cycle-wise earnings comparison
- ROI tracking (system-level)

10.12.3 User Analytics

- Top investors
- Top earners (profit + referral)
- Most active users
- Dormant accounts

10.12.4 Referral Analytics

- Referral tree depth distribution
- Conversion rate (referral → investment)
- Network expansion rate

10.13 Visualization Layer

Chart Types Used

- Line charts → growth trends
- Bar charts → comparisons
- Pie charts → distribution breakdown
- Tree views → referral structure

Dashboard Widgets

- Live investment counter
- Cycle countdown timer
- Profit distribution snapshot
- Referral activity heatmap (optional)

10.14 Data Filtering System

Global Filters

- Date range
- Cycle selection
- User ID
- Investment ID
- Admin ID

Advanced Filters

- Profit range filtering
- Referral level filtering
- Investment status filtering
- High-value user segmentation

10.15 Performance Optimization

Techniques Used

- Pre-aggregated reporting tables
- Cached analytics results
- Scheduled report generation
- Lazy loading for large datasets

10.16 Security & Access Control

Access Rules

- Super Admin → Full access
- Admin → Scoped access based on permissions
- User → Only personal reports

Protection

- Export logging
- Report access tracking
- Unauthorized access blocking

10.17 Output of Reports Module

The module produces:

- Financial statements (cycle-wise)
- User performance reports
- Investment summaries
- Referral network analytics
- Commission breakdown reports
- Audit compliance logs

11. Notifications System (Real-Time Alerts & Communication Engine)

The Notifications System is responsible for delivering real-time and scheduled updates across the Financial Referral CRM. It ensures that users, admins, and super admins are continuously informed about financial events, system actions, and cycle-based activities.

This module acts as the **communication backbone of the entire platform**.

11.1 Purpose of Notifications System

The system is designed to:

- Deliver real-time updates for financial events
- Notify users about investments, profits, and referrals
- Alert admins about system actions and approvals
- Support cycle-based automated messaging
- Maintain communication logs for audit purposes

11.2 Notification Channels

The system supports multiple delivery channels:

Primary Channels

- In-app notifications (mandatory)
- Email notifications (optional but recommended)

Optional Channels

- SMS alerts (configurable)
- WhatsApp API integration (future enhancement)

11.3 Notification Types

11.3.1 System Notifications

Triggered by system events:

- User registration success
- Investment creation confirmation
- System cycle start/end alerts

11.3.2 Financial Notifications

Triggered by financial engine:

- Profit calculated
- Profit cycle completed
- Referral commission generated
- Payout batch ready / released

11.3.3 Admin Notifications

Triggered for administrative actions:

- User modified
- Investment updated
- Payout released
- System configuration changes

11.3.4 Security Notifications

Triggered by security layer:

- Failed login attempts
- Suspicious activity detected
- Account blocked/unblocked
- Device/IP change alerts

11.4 Notification Flow Architecture

Event-Driven System

```
Event Trigger → Notification Engine → Channel Router → Delivery → Read Status
Tracking
```

Flow Steps

1. System event occurs (e.g., profit cycle completed)
2. Notification engine generates message
3. User role identified (User/Admin/Super Admin)
4. Channel selected (in-app/email/SMS)
5. Notification delivered
6. Read/unread status updated

11.5 User Notification Dashboard

Features

- Real-time notification feed
- Categorized alerts (Finance / System / Referral)
- Mark as read/unread
- Delete notification (optional)
- Notification history archive

Filtering Options

- Date range
- Notification type
- Priority level

11.6 Admin Notification Center

Features

- System-wide alerts
- User activity alerts
- Investment and payout alerts

- Approval requests (payout review, etc.)

Admin Actions via Notifications

- Approve requests directly
- View linked records
- Redirect to action panels

11.7 Super Admin Notifications

Critical Alerts

- System failures
- Financial mismatches
- Large payout events
- Security breaches
- Engine recalculation triggers

High Priority Rule

Critical alerts bypass all filters and appear instantly.

11.8 Notification Priority Levels

Level	Type	Behavior
High	Financial/Security	Instant push + email
Medium	System updates	In-app + email
Low	Informational	In-app only

11.9 Scheduling & Automation

Auto Notifications

- Cycle start reminders (1st, 15th)
- Cycle payout completion alerts
- Monthly summaries
- Investment reminders

Scheduled Jobs

- Daily system summary (optional)
- Weekly earnings summary
- Monthly performance report

11.10 Notification Data Structure

Each notification includes:

- Notification ID
- User ID
- Type (Finance/System/Security)
- Title
- Message
- Timestamp
- Read status
- Priority level
- Linked reference (Investment / User / Payout ID)

11.11 Real-Time Delivery System

Architecture

- WebSocket-based real-time push (recommended)
- Fallback polling system
- Queue-based message processing

Performance Strategy

- Batch notification processing for bulk events

- Rate limiting to avoid spam
- Deduplication of repeated alerts

11.12 Security Rules

- Notifications cannot be spoofed manually
- Only system or authorized admin can trigger alerts
- Sensitive financial notifications are encrypted in transit
- Audit log maintained for all notifications

11.13 Edge Cases Handling

11.13.1 Offline Users

- Notifications stored in queue
- Delivered upon next login

11.13.2 Duplicate Event Trigger

- System prevents duplicate notifications using event hash ID

11.13.3 Bulk Event Spike

- Queue throttling applied
- Priority-based delivery ensured

11.14 Notification Retention Policy

- Active notifications: stored indefinitely (or configurable)
- Archived notifications: moved after X days
- Audit logs: permanent retention

11.15 Output of Notifications System

The system generates:

- Real-time alerts
- Historical notification logs
- Delivery status tracking
- User engagement data

12. Database Schema (Relational Design – Core System Structure)

This section defines the complete relational database structure for the Financial Referral CRM. It is designed for **high consistency, auditability, and scalable financial computation**.

The schema is normalized but optimized with selective denormalization for reporting and performance.

12.1 Core Design Principles

- Fully relational structure (SQL-based recommended)
- Strong foreign key relationships
- Immutable financial records (ledger-style design)
- Audit-first architecture (no silent overwrites)
- Separation of:
  - Users
  - Investments
  - Referral tree
  - Commission ledger
  - Payout execution logs

12.2 Users Table

Table: `users`

Stores all user account information.

Fields

Field	Type	Description
id	UUID / BIGINT	Primary key
name	VARCHAR	Full name
email	VARCHAR	Unique email
phone	VARCHAR	Unique phone number
password_hash	TEXT	Encrypted password
referral_code	VARCHAR	Unique referral code
referred_by	UUID	Parent user ID
role	ENUM	USER / ADMIN / SUPER_ADMIN
status	ENUM	ACTIVE / BLOCKED / DELETED
created_at	TIMESTAMP	Registration time
updated_at	TIMESTAMP	Last update

Relations

- One user → many investments
- One user → many referrals (children)

12.3 Investments Table

Table: `investments`

Tracks all investment records.

Fields

Field	Type	Description
id	UUID	Primary key
user_id	UUID	Foreign key → users
amount	DECIMAL	Investment amount
start_date	DATE	Investment start
status	ENUM	ACTIVE / SUSPENDED / CLOSED
created_at	TIMESTAMP	Entry time

Relations

- One user → many investments

- One investment → many profit records

12.4 Profit Ledger Table

Table: `profit_ledger`

Stores all calculated profits per cycle.

Fields

Field	Type	Description
id	UUID	Primary key
investment_id	UUID	Foreign key
user_id	UUID	User reference
cycle_start	DATE	Cycle start
cycle_end	DATE	Cycle end
active_days	INT	Days counted
profit_amount	DECIMAL	Calculated profit
created_at	TIMESTAMP	Calculation time

12.5 Referral Tree Table

Table: `referrals`

Stores hierarchical relationships.

Fields

Field	Type	Description
id	UUID	Primary key
user_id	UUID	Child user
parent_id	UUID	Referrer
level	INT	Depth level
path	TEXT	Materialized path (e.g., A/B/C)
created_at	TIMESTAMP	Relationship creation

Key Concept

- Enables fast tree traversal
- Supports multi-level referral system

12.6 Commission Ledger Table

Table: `commission_ledger`

Stores all referral commissions (immutable financial record).

Fields

Field	Type	Description
id	UUID	Primary key
recipient_user_id	UUID	Who receives commission
source_user_id	UUID	Investor
investment_id	UUID	Investment reference
level	INT	Referral level
commission_amount	DECIMAL	Earned commission
status	ENUM	PENDING / CONFIRMED / REVERSED
created_at	TIMESTAMP	Entry time

Rule

- Never updated directly
- Only reversal entries allowed

12.7 Payout Batch Table

Table: `payout_batches`

Tracks system-generated payout groups.

Fields

Field	Type	Description
id	UUID	Batch ID
cycle_start	DATE	Cycle start
cycle_end	DATE	Cycle end
total_amount	DECIMAL	Total payout
status	ENUM	GENERATED / APPROVED / RELEASED
created_by	UUID	Super Admin ID
created_at	TIMESTAMP	Generation time

12.8 Payout Details Table

Table: `payout_details`

Stores per-user payout breakdown.

Fields

Field	Type	Description
id	UUID	Primary key
batch_id	UUID	Foreign key
user_id	UUID	Recipient
amount	DECIMAL	Payout amount
type	ENUM	PROFIT / COMMISSION
status	ENUM	PENDING / RELEASED
created_at	TIMESTAMP	Entry time

12.9 Admin Activity Log Table

Table: `admin_logs`

Tracks all admin actions.

Fields

Field	Type	Description
id	UUID	Primary key
admin_id	UUID	Admin user
action	VARCHAR	Action type

Field	Type	Description
target_type	VARCHAR	USER / INVESTMENT / SYSTEM
target_id	UUID	Affected record
metadata	JSON	Extra details
ip_address	VARCHAR	Security tracking
created_at	TIMESTAMP	Timestamp

12.10 System Audit Log Table

Table: `audit_logs`

Stores system-level events (immutable).

Fields

Field	Type	Description
id	UUID	Primary key
event_type	VARCHAR	ENGINE / CALCULATION / SECURITY
reference_id	UUID	Related entity
details	JSON	Full event data
created_at	TIMESTAMP	Timestamp

12.11 Notifications Table

Table: `notifications`

Stores all system notifications.

Fields

Field	Type	Description
id	UUID	Primary key
user_id	UUID	Recipient
type	ENUM	SYSTEM / FINANCE / SECURITY
title	VARCHAR	Notification title
message	TEXT	Content
priority	ENUM	HIGH / MEDIUM / LOW
read_status	BOOLEAN	Seen or not

Field	Type	Description
created_at	TIMESTAMP	Timestamp

12.12 System Settings Table

Table: `system_settings`

Stores configurable system rules.

Fields

Field	Type	Description
id	UUID	Primary key
key	VARCHAR	Setting name
value	TEXT	Value
updated_by	UUID	Admin ID
updated_at	TIMESTAMP	Timestamp

12.13 Relationships Overview

Core Relations Map

- users → investments (1:N)
- users → referrals (1:N self-reference)
- investments → profit_ledger (1:N)
- users → commission_ledger (1:N)
- payout_batches → payout_details (1:N)
- users → notifications (1:N)

12.14 Indexing Strategy (Performance Critical)

Recommended Indexes

- users(referral_code)
- users(referred_by)

- investments(user_id, status)
- profit_ledger(cycle_start, cycle_end)
- commission_ledger(recipient_user_id)
- payout_details(batch_id, user_id)

12.15 Data Integrity Rules

- Foreign key constraints enforced
- No orphan referral records
- No duplicate payout entries per user per batch
- Immutable ledger tables (profit + commission + audit)

12.16 Scalability Enhancements

- Partition `profit_ledger` by month
- Partition `commission_ledger` by cycle
- Archive old `audit_logs`
- Use read replicas for reporting module

12.17 Output of Database Design

This schema supports:

- Investment tracking system
- Cycle-based profit engine
- Multi-level referral system
- Commission ledger system
- Payout execution system
- Full audit compliance system
- High-scale reporting & analytics

13. API Specification (Complete Backend Endpoints Design)

This section defines the full REST API layer for the Financial Referral CRM. It is structured for **scalable backend development**, role-based access control, and clean separation of modules.

All APIs follow:

- Base URL: `/api/v1`
- Auth: JWT-based authentication
- Role-based access: USER / ADMIN / SUPER_ADMIN
- Response format: JSON
- Standard error handling across all endpoints

13.1 Authentication APIs

POST `/auth/register`

Create new user

Body

```
{
  "name": "John Doe",
  "email": "john@example.com",
  "phone": "9999999999",
  "password": "******",
  "referral_code": "ABC123"
}
```

Response

- User created
- Referral linked (if valid code)

POST `/auth/login`

User login

Response

- JWT token
- User role
- Profile info

POST `/auth/logout`

Invalidate session

GET `/auth/profile`

Get logged-in user profile

13.2 User Management APIs (Admin/Super Admin)

GET `/users`

List all users (paginated)

Query Params

- status
- role
- search
- page

POST `/users`

Create user (Super Admin only)

PUT `/users/:id`

Update user details

DELETE `/users/:id`

Delete user (Super Admin only)

PATCH `/users/:id/block`

Block user

PATCH `/users/:id/unblock`

Unblock user

GET `/users/:id/referrals`

Get referral tree for a user

13.3 Investment APIs

POST `/investments`

Create investment

Body

```
{
  "user_id": "UUID",
  "amount": 10000,
  "start_date": "2026-06-01"
}
```

GET `/investments`

List investments (filters supported)

Query Params

- user_id
- status
- date range

GET `/investments/:id`

Investment details

PATCH `/investments/:id`

Update investment (Admin/Super Admin)

DELETE `/investments/:id`

Remove investment (Super Admin only)

13.4 Profit Engine APIs

POST `/profit/calculate-cycle`

Trigger profit calculation manually

Body

```
{
  "cycle_start": "2026-06-01",
  "cycle_end": "2026-06-14"
}
```

GET `/profit/cycles`

List all profit cycles

GET `/profit/user/:id`

User profit history

POST `/profit/recalculate`

Re-run profit engine (Super Admin only)

13.5 Referral APIs

GET `/referrals/tree/:userId`

Get referral tree structure

GET `/referrals/summary/:userId`

Referral statistics

POST `/referrals/assign`

Manually assign referral (Super Admin only)

GET `/referrals/network`

Full referral network overview (Admin/Super Admin)

13.6 Commission APIs

GET `/commissions/user/:userId`

User commission history

GET `/commissions/ledger`

Full commission ledger (Admin/Super Admin)

GET `/commissions/summary`

Aggregated commission analytics

POST `/commissions/recalculate`

Recalculate commissions (Super Admin only)

13.7 Payout APIs (NO WALLET SYSTEM)

POST `/payout/generate`

Generate payout batch for cycle

GET `/payout/batches`

List all payout batches

GET `/payout/batch/:id`

Batch details

POST `/payout/release/:batchId`

One-click payout release (Super Admin only)

GET `/payout/user/:userId`

User payout history

13.8 Reports APIs

GET `/reports/users`

User report

GET `/reports/investments`

Investment report

GET `/reports/profit`

Profit report

GET `/reports/referrals`

Referral analytics report

GET `/reports/commissions`

Commission report

GET `/reports/payouts`

Payout report

GET `/reports/audit`

Audit logs report

GET `/reports/export`

Export reports

Query Params

- format: pdf | excel | csv | json
- type: report type
- date range

13.9 Notification APIs

GET `/notifications`

Get user notifications

PATCH `/notifications/:id/read`

Mark notification as read

POST `/notifications/send`

Send system notification (Admin/Super Admin)

13.10 Admin APIs

GET `/admin/dashboard`

Admin dashboard data

GET `/admin/activity-logs`

Admin activity logs

POST `/admin/create`

Create admin account (Super Admin only)

PATCH `/admin/permissions/:id`

Update admin permissions

13.11 Super Admin System APIs

GET `/system/settings`

Get system configuration

PATCH `/system/settings`

Update system rules (profit %, cycles, etc.)

GET `/system/health`

System status monitoring

POST `/system/override`

Manual override for financial corrections

13.12 Security APIs

GET `/security/logins`

Login history tracking

GET `/security/alerts`

Security alerts

POST `/security/lock-user/:id`

Emergency user lock

13.13 API Security Rules

- JWT authentication required for all protected routes
- Role-based middleware enforcement
- Rate limiting on sensitive APIs
- Audit logging for all financial endpoints
- No direct DB access allowed from frontend

13.14 API Response Standard

Success Response

```
{
  "status": "success",
  "data": {},
  "message": "Operation completed"
}
```

Error Response

```
{
  "status": "error",
  "message": "Invalid request",
  "code": 400
}
```

13.15 System Behavior Summary

This API layer supports:

- Full CRM operations
- Investment lifecycle management
- Profit engine execution
- Referral and commission tracking
- One-click payout system
- Complete reporting suite
- System administration and security control

14. Wireframes (UI/UX Structure & Screen Layout Design)

This section defines the visual structure and screen-level layout of the Financial Referral CRM system. These wireframes are **functional blueprints**, not visual designs, and are meant to guide frontend development.

The system is divided into three main interfaces:

- Super Admin Panel
- Admin Panel
- User (Investor) Dashboard

14.1 Super Admin Dashboard Wireframe

Layout Structure

```
---------------------------------------------------------
| TOP BAR: Logo | Search | Notifications | Profile      |
---------------------------------------------------------
| SIDEBAR:                                            |
| - Dashboard                                         |
| - Users (CRUD)                                      |
| - Investments                                       |
| - Profit Engine                                   |
```

```
| - Referral Tree                                    |
| - Commission Engine                                |
| - Payout System                                    |
| - Reports                                          |
| - Admin Management                                 |
| - System Settings                                 |
---------------------------------------------------------
| MAIN DASHBOARD AREA                                |
---------------------------------------------------------
```

Dashboard Widgets (Main Area)

- Total Users Card
- Active Investments Card
- Total Profit Generated Card
- Total Payout Released Card
- System Liability Card

Quick Actions Panel

- Create User
- Create Investment
- Run Profit Cycle
- Generate Payout Batch
- Release Payout (One Click Button)

Live Activity Feed

- New registrations
- Investments created
- Payout executed
- Admin actions log

14.2 User Management Screen (Super Admin)

Layout

```
---------------------------------------------------------
| Filters: Search | Status | Role | Date Range          |
---------------------------------------------------------
```

```
| USER TABLE                                            |
---------------------------------------------------------
| ID | Name | Email | Ref Code | Status | Actions      |
---------------------------------------------------------
```

User Detail Drawer (Right Panel)

- Profile Info
- Referral Tree View (expandable)
- Investments List
- Profit History
- Commission History

14.3 Investment Management Screen

Layout

```
---------------------------------------------------------
| Filters: User | Status | Date Range                    |
---------------------------------------------------------
| INVESTMENT TABLE                                      |
---------------------------------------------------------
| ID | User | Amount | Start Date | Status | Actions   |
---------------------------------------------------------
```

Investment Detail View

- Investment timeline
- Cycle participation
- Profit earned per cycle

14.4 Profit Engine Screen

Layout

```
---------------------------------------------------------
| Cycle Selector | Run Calculation | Recalculate         |
---------------------------------------------------------
| CYCLE STATUS PANEL                                    |
---------------------------------------------------------
| Cycle: 1-14 / 15-End                                  |
| Total Profit Generated                                 |
```

```
| Active Investments Processed                          |
---------------------------------------------------------
```

Batch Output Table

- Investment ID
- User ID
- Profit amount
- Active days
- Status

14.5 Referral Tree Visualization Screen

Layout

```
---------------------------------------------------------
| Search User                                            |
---------------------------------------------------------
| TREE VISUALIZATION AREA                                |
---------------------------------------------------------
|        USER A                                        |
|       /      \                                       |
|   USER B     USER C                                 |
|   /   \         \                                    |
| USER D USER E   USER F                              |
---------------------------------------------------------
```

Controls

- Expand / collapse nodes
- Zoom in/out
- Level filter (1–5 depth)

14.6 Commission Engine Screen

Layout

```
---------------------------------------------------------
| Filters: User | Level | Date Range                      |
---------------------------------------------------------
| COMMISSION LEDGER TABLE                               |
---------------------------------------------------------
```

```
| User | Source | Level | Amount | Status | Date     |
---------------------------------------------------------
```

Summary Panel

- Total Commission Paid
- Level-wise distribution chart
- Top earners list

14.7 Payout System Screen (NO WALLET MODEL)

Layout

```
---------------------------------------------------------
| Cycle Selector | Generate Batch | Release All        |
---------------------------------------------------------
| PAYOUT BATCH TABLE                                   |
---------------------------------------------------------
| Batch ID | Cycle | Total Amount | Status | Action   |
---------------------------------------------------------
```

Batch Detail View

- User-wise payout breakdown
- Total payout summary
- One-click release button

14.8 Reports Dashboard Screen

Layout

```
---------------------------------------------------------
| Report Type Selector | Date Range | Export Button    |
---------------------------------------------------------
| REPORT VIEW AREA                                     |
---------------------------------------------------------
| Charts + Tables                                      |
---------------------------------------------------------
```

Available Tabs

- Users
- Investments
- Profit
- Referrals
- Commissions
- Payouts
- Audit Logs

14.9 Admin Panel Wireframe

Layout

```
---------------------------------------------------------
| Dashboard | Users | Investments | Reports | Profile   |
---------------------------------------------------------
| SUMMARY WIDGETS                                     |
---------------------------------------------------------
| Assigned Users | Investments | Pending Actions       |
---------------------------------------------------------
```

Admin User Table Screen

- Limited user access view
- No delete permissions (optional)
- Status updates only

Admin Investment Screen

- View assigned investments
- Verify entries
- Flag issues

14.10 User Dashboard Wireframe

Layout

```
---------------------------------------------------------
| Top Bar: Profile | Notifications | Logout             |
---------------------------------------------------------
```

```
| SIDEBAR:                                               |
| - Dashboard                                            |
| - Investments                                          |
| - Profit History                                     |
| - Referral Tree                                      |
| - Transactions                                       |
----------------------------------------------------------
```

Main Dashboard Widgets

- Total Investment
- Total Profit Earned
- Referral Earnings
- Active Cycle Status

Investment Screen

- Investment list
- Profit per investment
- Cycle tracking

Referral Screen

- Referral code display
- Referral tree view
- Earnings from referrals

Profit Screen

- Cycle-wise breakdown
- Total earnings chart
- Historical profit table

14.11 UI Behavior Rules

- All tables are paginated
- All financial actions require confirmation modal

- One-click payout requires double confirmation
- Real-time updates via WebSocket
- Role-based UI rendering (dynamic menus)

14.12 UX Flow Summary

User Flow:

Login → Dashboard → Investments → Profit → Referrals → History

Admin Flow:

Dashboard → Users → Investments → Reports → Approval Tasks

Super Admin Flow:

Full system access → Financial control → Payout execution → System configuration

---

14.13 Frontend Tech Recommendation

- React / Next.js (recommended)
- TailwindCSS for UI
- Chart.js or Recharts for analytics
- WebSocket for live updates
- Role-based route protection

14.14 Output of Wireframes Module

This section defines:

- Complete screen structure
- Navigation flow
- UI component layout
- Role-based dashboards
- Functional interaction points

15. Security (Authentication, Encryption, Fraud Detection & System Hardening)

The Security module defines the complete protection framework for the Financial Referral CRM. Since the system handles financial calculations, referrals, and payout execution, security is treated as a **core architectural layer**, not an add-on.

15.1 Security Objectives

The system is designed to ensure:

- Only authorized users access the platform
- Financial data cannot be tampered with
- All transactions are fully traceable
- Fraudulent activity is detected early
- API endpoints are protected from abuse
- System integrity is preserved under load or attack

15.2 Authentication System

15.2.1 JWT-Based Authentication

- All sessions are authenticated using JWT tokens
- Tokens include:
    - user_id
    - role (USER / ADMIN / SUPER_ADMIN)
    - session expiry

15.2.2 Login Security Rules

- Maximum failed login attempts: configurable (e.g., 5)
- Account lockout after repeated failures
- IP tracking for every login attempt
- Device fingerprint logging (optional enhancement)

15.2.3 Multi-Role Access Control (RBAC)

Role	Access Level
User	Own data only
Admin	Scoped operational access
Super Admin	Full system control

15.3 Password & Credential Security

Password Rules

- Stored using strong hashing (bcrypt / Argon2)
- No plaintext passwords stored anywhere
- Minimum complexity enforcement:
  - Uppercase + lowercase + number + symbol

Reset Mechanism

- Secure token-based password reset
- Expiry time enforced (e.g., 10–15 minutes)
- Single-use reset links only

15.4 Data Encryption Layer

15.4.1 Data in Transit

- HTTPS (TLS 1.2 or higher)
- All API communication encrypted

15.4.2 Data at Rest

- Sensitive fields encrypted:
  - passwords
  - financial amounts (optional)
  - API keys

- Database-level encryption supported

15.4.3 Key Management

- Centralized secret management system
- Rotation policy for encryption keys
- No hardcoded secrets in codebase

15.5 API Security Layer

15.5.1 Authentication Middleware

- Every request validated via JWT
- Token expiry enforced

15.5.2 Rate Limiting

- Prevent API abuse
- Example limits:
    - 100 requests/min per user
    - stricter limits on financial endpoints

15.5.3 Role-Based Endpoint Protection

- User APIs → restricted to self-data
- Admin APIs → scoped access only
- Super Admin APIs → full privileges

15.5.4 Input Validation

- Strict schema validation (DTO layer)
- SQL injection prevention
- XSS protection
- No raw query execution from frontend

15.6 Fraud Detection System

15.6.1 Investment Anomaly Detection

Flags:

- Multiple investments in short time window
- Unusual large amount spikes
- Repeated pattern investments

15.6.2 Referral Fraud Detection

- Self-referral attempts → blocked
- Circular referral detection → blocked
- Fake referral chain patterns → flagged

15.6.3 Commission Abuse Detection

- Duplicate commission generation attempts
- Overlapping level payout inconsistencies
- Unexpected commission spikes

15.6.4 Behavior Monitoring

- Unusual login patterns
- Multiple IP changes in short time
- Bot-like request behavior

15.7 Audit Integrity System

15.7.1 Immutable Logs

- Profit ledger is read-only

- Commission ledger is append-only
- Audit logs cannot be edited or deleted

15.7.2 Event Tracking

Every critical event is logged:

- User creation/modification
- Investment creation/update
- Profit calculation runs
- Commission generation
- Payout execution
- Admin overrides

15.7.3 Audit Trail Structure

Each log contains:

- Actor (User/Admin/System)
- Action type
- Timestamp
- Before/After state (if applicable)
- IP address
- Device info

15.8 System Hardening

15.8.1 Infrastructure Security

- Firewall protection
- DDoS mitigation layer
- API gateway enforcement
- Secure hosting environment

15.8.2 Database Security

- Role-based DB access

- Read/write separation
- Query monitoring enabled
- Backup encryption

15.8.3 Server Hardening

- Disable unnecessary ports
- Secure SSH access
- Regular patch updates
- Container isolation (if using Docker)

15.8.4 Session Security

- Auto logout on inactivity
- Single session enforcement (optional)
- Session invalidation on password change

15.9 Financial Security Rules

15.9.1 Payout Protection

- One-click payout requires Super Admin confirmation
- Payout batch is locked before release
- No modification after approval

15.9.2 Calculation Protection

- Profit engine is deterministic (no manual randomness)
- Rate changes apply only to future cycles
- Recalculation is logged and versioned

15.9.3 Double-Spend Prevention

- No duplicate payout entries allowed

- Unique transaction IDs enforced
- Ledger validation before execution

15.10 Monitoring & Alerting System

Real-Time Alerts

- Failed login spikes
- Fraud detection triggers
- Large payout execution alerts
- System anomalies

Admin Alerts

- Suspicious user activity
- Investment anomalies
- Commission mismatches

Super Admin Alerts

- System integrity breaches
- Engine failures
- Critical financial mismatches

15.11 Security Compliance Layer

- Full audit traceability
- Role-based access enforcement
- Financial immutability compliance
- Tamper-proof logs
- Exportable audit reports

15.12 Penetration Resistance Features

- Brute force protection
- API throttling
- Payload sanitization
- Request signature validation (optional enhancement)

15.13 Security Summary

This system ensures:

- No unauthorized financial manipulation
- Fully traceable investment and payout flow
- Fraud detection at multiple layers
- Strong API and database protection
- Immutable financial audit history
- Enterprise-grade system hardening

16. Deployment (Architecture, Hosting, CI/CD, Scaling & Production Rollout)

This section defines how the Financial Referral CRM is deployed, scaled, and maintained in production. The goal is to ensure **high availability, zero data loss, and smooth financial cycle execution**.

16.1 Deployment Architecture Overview

The system follows a **modular cloud-based architecture**:

- Frontend: Web dashboard (User / Admin / Super Admin)
- Backend: REST API service layer
- Database: Relational SQL database
- Background Workers: Profit engine + commission engine
- Scheduler: Cycle automation (14th & end of month)
- Cache Layer: Performance optimization (Redis recommended)

16.2 Recommended Tech Stack

Frontend

- React / Next.js
- TailwindCSS
- Chart.js / Recharts

Backend

- Node.js (NestJS / Express) or Java Spring Boot

Database

- PostgreSQL (recommended)
- MySQL (alternative)

Infrastructure

- AWS / Azure / Google Cloud

Background Jobs

- BullMQ / RabbitMQ / Kafka (for financial engines)

16.3 Environment Structure

Environments

- Development (DEV)
- Staging (UAT)
- Production (LIVE)

Environment Rules

- DEV → testing only
- UAT → admin approval testing
- PROD → live financial system (locked)

16.4 CI/CD Pipeline

Pipeline Flow

```
Code Push → Build → Test → Security Scan → Deploy → Health Check
```

Automation Steps

- Linting + validation
- Unit tests
- API contract tests
- Database migration checks
- Deployment to staging first
- Manual approval → production release

16.5 Database Deployment Strategy

Schema Migration

- Version-controlled migrations
- Rollback support for failed deployments
- Zero-downtime migration preferred

Backup Strategy

- Daily automated backups
- Point-in-time recovery (PITR)
- Encrypted backup storage

16.6 Backend Deployment

Service Structure

- API Server (Core CRM)
- Worker Service (Profit + Commission Engine)

- Scheduler Service (Cycle automation)

Scaling Model

- Horizontal scaling for API servers
- Worker queue scaling for heavy financial loads
- Load balancer in front of APIs

16.7 Frontend Deployment

Hosting Options

- Vercel (recommended for Next.js)
- AWS S3 + CloudFront
- Netlify (alternative)

Performance Rules

- Lazy loading for dashboards
- Code splitting per role (User/Admin/Super Admin)
- CDN caching enabled

16.8 Background Job System

Core Workers

- Profit Calculation Worker
- Commission Calculation Worker
- Payout Batch Generator
- Notification Worker

Queue System

- Priority queues:
  - HIGH → financial calculations
  - MEDIUM → reports
  - LOW → notifications

16.9 Scheduling System

Cycle Automation

- Runs automatically on:
  - 14th of every month
  - Last day of every month

Scheduler Rules

- Idempotent execution (no duplicate runs)
- Retry mechanism for failures
- Logging for every cycle execution

16.10 Scaling Strategy

Horizontal Scaling

- Multiple API instances behind load balancer
- Stateless backend design

Database Scaling

- Read replicas for reports
- Write master for transactions
- Partitioning for large ledger tables

Worker Scaling

- Auto-scale workers based on queue load

16.11 Performance Optimization

- Redis caching for:
    - Referral tree lookups
    - Dashboard metrics
- Precomputed analytics tables
- Indexed financial tables
- Batch processing for large datasets

16.12 Monitoring & Logging

System Monitoring

- CPU / Memory tracking
- API latency tracking
- Error rate monitoring

Financial Monitoring

- Profit engine execution logs
- Commission calculation logs
- Payout execution tracking

Tools

- Prometheus + Grafana
- ELK Stack (Elasticsearch, Logstash, Kibana)

16.13 Security in Deployment

- HTTPS enforced everywhere

- API gateway protection
- WAF (Web Application Firewall)
- DDoS protection layer
- Environment-based secret management

16.14 Rollback Strategy

If deployment fails:

- Automatic rollback to previous stable version
- Database rollback via migration versioning
- Worker queue pause during rollback

16.15 High Availability Design

- Multi-zone deployment
- Failover database setup
- Redundant API servers
- Backup scheduler instance

16.16 Disaster Recovery Plan

Recovery Steps

- Restore latest backup
- Replay transaction logs
- Re-run financial engine safely
- Validate audit consistency

RTO / RPO Targets

- RTO (Recovery Time): < 1 hour
- RPO (Data Loss): near zero (transaction-safe backups)

16.17 Production Deployment Flow

1. Code freeze
2. Staging validation
3. Financial engine dry-run
4. Admin approval
5. Production deployment
6. Health check validation
7. System unlock

16.18 Deployment Summary

This deployment architecture ensures:

- Stable financial cycle execution
- Zero-downtime upgrades (where possible)
- Scalable referral + investment system
- Safe profit and payout processing
- Enterprise-grade reliability

17. Future Enhancements (Roadmap, AI Expansion & System Evolution)

This section defines the long-term evolution plan for the Financial Referral CRM. It focuses on improving scalability, intelligence, automation, and transparency while maintaining the core financial engine integrity.

17.1 AI-Based Fraud Detection System

Objective

Upgrade current rule-based fraud detection into a **machine learning-powered anomaly detection system**.

Capabilities

- Detect unusual investment patterns
- Identify referral farming behavior

- Flag abnormal commission spikes
- Predict potential system abuse before it happens

AI Models

- Classification model (fraud / non-fraud)
- Anomaly detection (unsupervised learning)
- Behavioral clustering (user profiling)

Data Inputs

- Investment frequency
- Referral velocity
- Login behavior
- Transaction patterns
- Device/IP history

17.2 Blockchain-Based Ledger System (Optional Upgrade)

Objective

Introduce an immutable distributed ledger for financial transparency.

Use Cases

- Investment records on-chain hash storage
- Profit cycle verification
- Commission transparency proof
- Payout audit trail validation

Implementation Approach

- Store hashes of:
    - profit_ledger entries
    - commission_ledger entries
    - payout batches
- Anchor them to blockchain (Ethereum / Polygon / private chain)

Benefit

- Tamper-proof financial records
- External audit verification
- Increased trust in system integrity

17.3 Mobile Application (Android & iOS)

Objective

Extend CRM access to mobile users and investors.

Features

User App

- Investment tracking
- Profit updates in real-time
- Referral tree view
- Notifications
- Earnings dashboard

Admin App (Optional)

- Approve payouts
- Monitor system activity
- View reports on mobile

Tech Stack

- Flutter (recommended)
- React Native (alternative)

17.4 Advanced Analytics Engine

Objective

Transform raw financial data into predictive insights.

Capabilities

- Profit forecasting per cycle
- User lifetime value prediction
- Referral growth forecasting
- Investment trend analysis

Dashboards

- ROI heatmaps
- Growth trajectory graphs
- Network expansion visualization

17.5 Automation Upgrades

Current System

- Rule-based profit cycles
- Manual payout release

Future Automation

- Auto payout approval based on thresholds
- Smart cycle execution (AI-triggered optimization)
- Auto risk blocking for suspicious users
- Dynamic commission adjustment engine

Smart Rules Engine

- If risk score > threshold → auto freeze
- If payout variance is normal → auto-approve batch (optional)
- If system load is high → defer non-critical tasks

17.6 Real-Time Streaming Architecture Upgrade

Enhancement

Move from batch-based updates → real-time financial streaming

Features

- Live profit updates per investment
- Real-time referral commission crediting
- Instant dashboard refresh
- Streaming audit logs

17.7 Scalability Roadmap

Phase 1 (Current System)

- Monolithic or modular backend
- SQL database
- Batch processing engine

Phase 2 (Scaling)

- Microservices architecture:
    - User Service
    - Investment Service
    - Profit Engine Service
    - Referral Service
- Queue-based processing (Kafka/RabbitMQ)

Phase 3 (Enterprise Scale)

- Distributed system architecture
- Multi-region deployment
- Auto-scaling workers
- Global CDN distribution

17.8 Performance Enhancements

Database Level

- Sharding for large user base
- Partitioning of ledger tables
- Read replicas for analytics

Backend Level

- Async processing for all financial operations
- Caching layer for referral trees
- Precomputed dashboards

Frontend Level

- Server-side rendering (SSR)
- Lazy-loaded dashboards
- Real-time WebSocket updates

17.9 Advanced Reporting & BI Layer

Future Upgrade

Integrate Business Intelligence tools:

- Power BI / Tableau integration
- Custom BI dashboard inside CRM
- Predictive financial modeling reports

17.10 Smart Notification System (AI-Driven)

Enhancement

Instead of static alerts:

- Personalized notifications
- Behavioral-triggered alerts
- Smart engagement reminders

Examples

- “Your referral earnings increased by 12% this week”
- “Unusual investment pattern detected in your network”

17.11 API Evolution Roadmap

Versioning Strategy

- v1 → current system APIs
- v2 → microservices split APIs
- v3 → AI + blockchain integrated APIs

Future API Features

- GraphQL layer (optional)
- Webhook system for external integrations
- Public API for analytics partners

17.12 Security Enhancements (Future)

- Zero-trust architecture
- Behavioral authentication (AI-based login validation)
- Biometric login for mobile apps
- Advanced threat detection systems

17.13 Global Expansion Readiness

Multi-Currency Support

- INR, USD, EUR support
- Dynamic conversion layer

Multi-Timezone System

- Region-based cycle calculation support

Localization

- Multi-language UI support

17.14 Ecosystem Expansion

Future modules can include:

- Loan management system
- Staking-based investment models
- Marketplace integration
- Partner CRM system

17.15 Long-Term Vision Summary

The Financial Referral CRM evolves into:

- A **fully automated financial ecosystem**
- AI-driven fraud-proof investment platform
- Globally scalable referral + investment network
- Transparent (optionally blockchain-backed) financial ledger system
- Mobile-first financial tracking platform